Investigation of Intermediates in the Reaction of Chenodeoxycholic Acid with Sulfuric Acid Different Wavelengths by Comparing the Activation Energies

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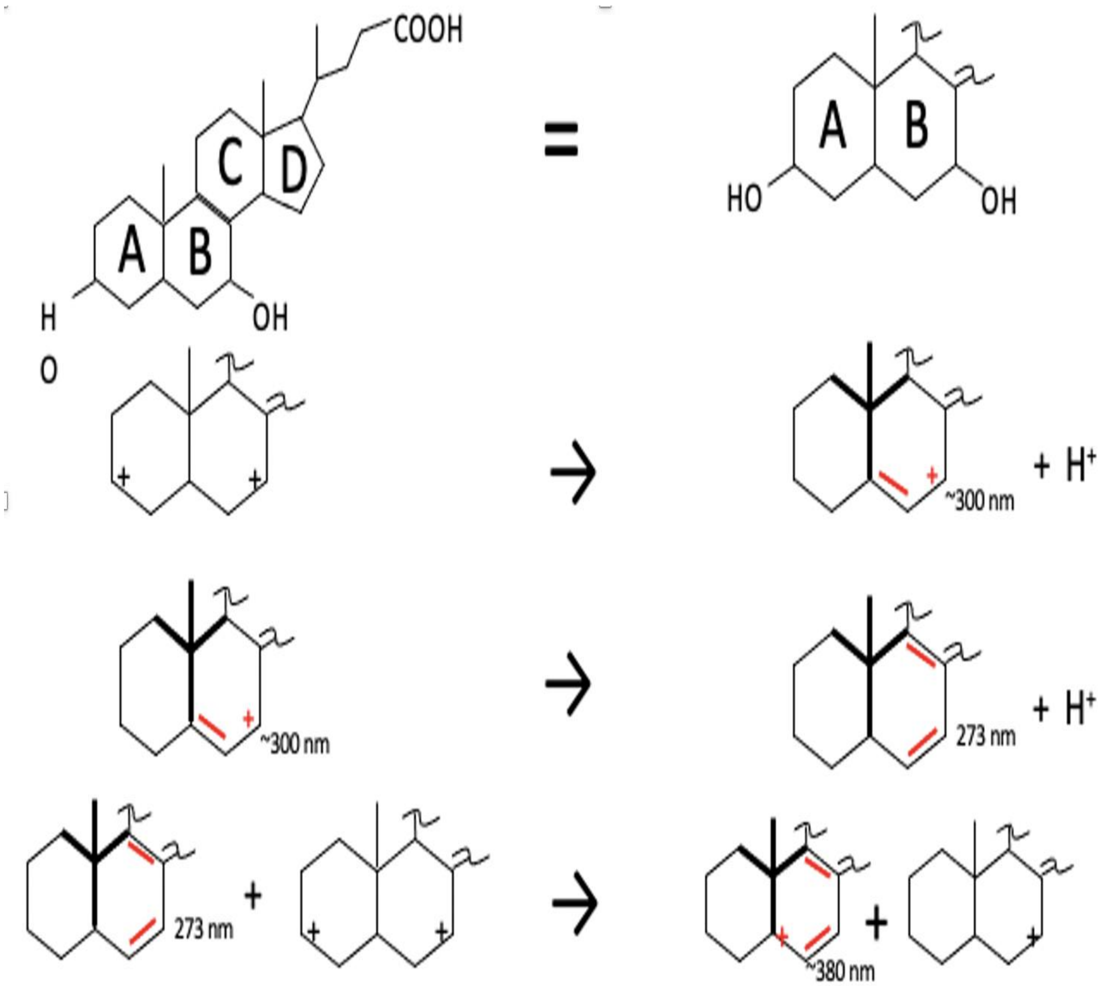
Abstract

0.01 mg chenodeoxycholic acid was used to obtain the activation energy at different temperatures with 74% w/w H_2SO_4 by UV-visible spectra to compare activation energies at different wavelengths. UV-visible spectra were used to collect the data at 50,55,60 and 65 °C, and the rate constant and reaction order were found by Guggenheim treatment. Moreover, an Arrhenius plot was constructed to find the activation energies at different wavelengths. The wavelength 263,310 and 380 nm were selected to show various activation energies at four different temperatures. The activation energies were 43.4 kJ/mol at 263 nm, 20.8 kJ/mol at 310 nm, and 90.2 kJ/mol at 380 nm. For future work, the correlation between temperature and activation energy can be studied to determine the optimal condition(temperature and concentration of sulfuric acid) of the chenodeoxycholic acid reaction with sulfuric acid.

Introduction

Chenodeoxycholic acid is a bile acid naturally found in the body. It works by dissolving the cholesterol that makes gallstones and inhibiting cholesterol production in the liver, and absorption in the intestines. In this experiment, the Guggenheim analysis method (($\ln[C]_{i+1}$ - $\ln[C]_{i-1}$vs time) and the Arrhenius equation ($k=Ae^{-Ea/RT}$) determine the reaction order, the rate constant, and the activation energy at different wavelengths. The purpose of this research is to observe the various activation energies of chenodeoxycholic acid at different wavelengths. When 0.05g of chenodeoxycholic acid reacts with 74% w/w sulfuric acid at four different temperatures (50,55,60,65 °C), it forms the intermediates and products: allylic cation and dienylic cation. Figure 1 shows the proposed mechanism for this reaction. The predicted UV-visible spectra range of the Chenodeoxycholic acid is around 270 nm, allylic action is around 300 nm, and dienylic cation is around 380 nm.



(1)

(2)

(3)

Figure 1. Proposed mechanism of chenodeoxycholic acid with hydride ion

The Arrhenius equation finds the activation energies at each wavelength by using slope of the natural log of the observed rate constant and inverse of the kelvin temperature. The equation 1 shows the relationship of all these variables. In this equation, k is the rate constant, Ea is the activation energy, R is the gas constant, and T is the temperature in kelvin. The greater slope indicates the higher the activation energy.

$$\ln k = lnA - \frac{Ea}{RT} \tag{1}$$

Method

The Active temperature control unit (FisherScientific, isotemp 3016) was set to 50,55,60,65° C at different times, and the Activate spectrophotometer (Shimadzu UV-visible spectrophotometer, UV-2600) was connected to UVProbe v.252 software and the temperature control unit. The spectrophotometer parameters were measured from 200 to 440nm wavelength range, fast scan speed, 0.7 nm slit, repeat scan, 76 seconds between scans. 74% of 3.5 mL of the H2SO 4 catalyst was added to both the reaction and reference cuvettes (Starna Spectrosil, 1-Q-10-GL14C) by a 10 mL grad pipet with a (green) thumbwheel suction and delivery device. When the temperature control unit had reached the setpoint temperature, the BASELINE was measured on the machine with the cell-door closed and no cuvettes in the instrument. The cuvettes were placed in the cell holders with frosted sides facing the operator. The reaction cuvette was placed front, and the reference cuvette was placed rear side. After solutions had reached thermal equilibrium, the cell compartment was closed. The Hamilton 780 model GC syringe 50 microliter capacity syringe was used to inject the calculated amount of chenodeoxycholic acid (in methanol solvent) into the reaction (front) cell after pressing the AUTOZERO button. When the calculated L volume of chenodeoxycholic acid was loaded into a syringe, the START button was pressed on the spectrophotometer, and the cap on the front cuvette was removed. The calculated amount of chenodeoxycholic acid was immediately injected, and the cap was quickly recapped. The cuvette was inverted twice and replaced, and the observation had started when the lid was closed. When all the data was collected, it was transferred to an Excel file for the calculation part.

Results

Figure 2, 3, 4 and 5 shows the absorbance vs wavelength graph of chenodeoxycholic acid with 74% w/w H2SO4 at 50,55,60 and 65°C. The smooth curves with no point indicates that the λ_{max} is around 380nm.

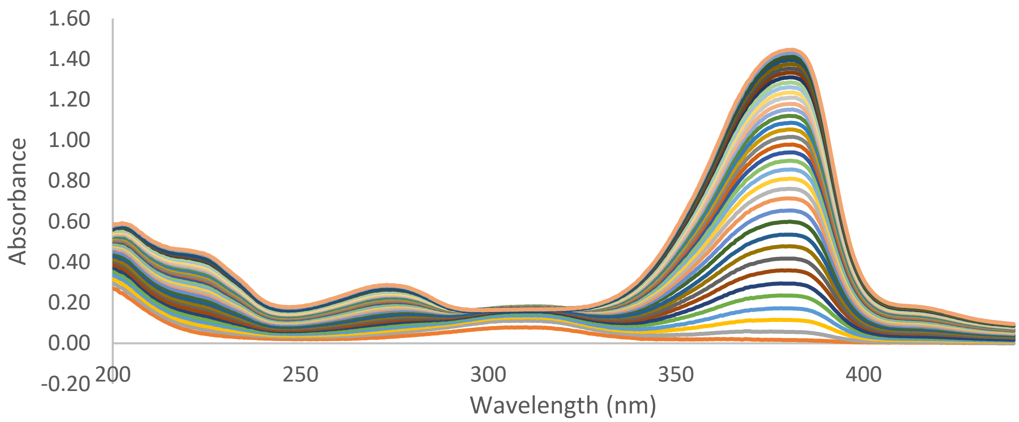


Figure 2. Time-series absorbance spectra at 50°C for the reaction of 0.01mg chenodeoxycholic acid with 74% w/w H2SO4. The observed maximum absorbance appears at approximately 380 nm.

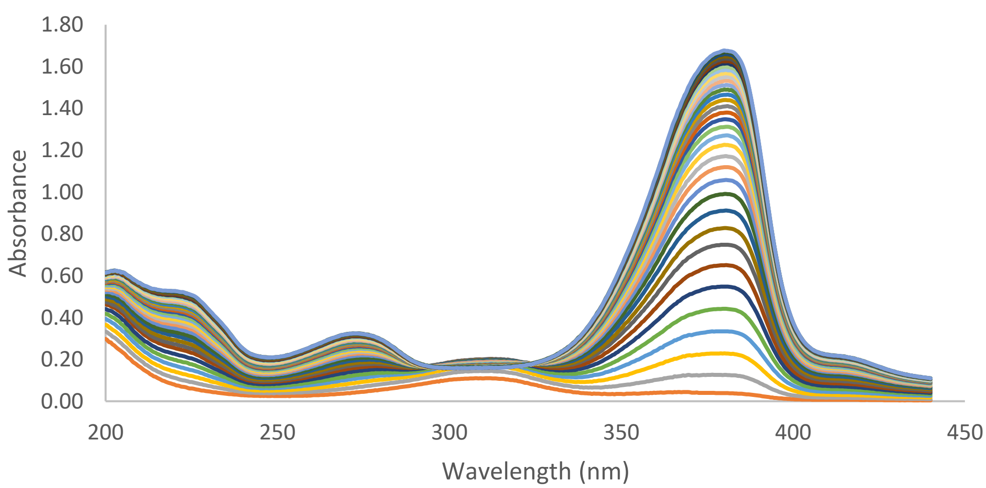


Figure 3. Time-series absorbance spectra at 55°C for the reaction of 0.01mg chenodeoxycholic acid with 74% w/w H2SO4. The observed maximum absorbance appears at approximately 380 nm.

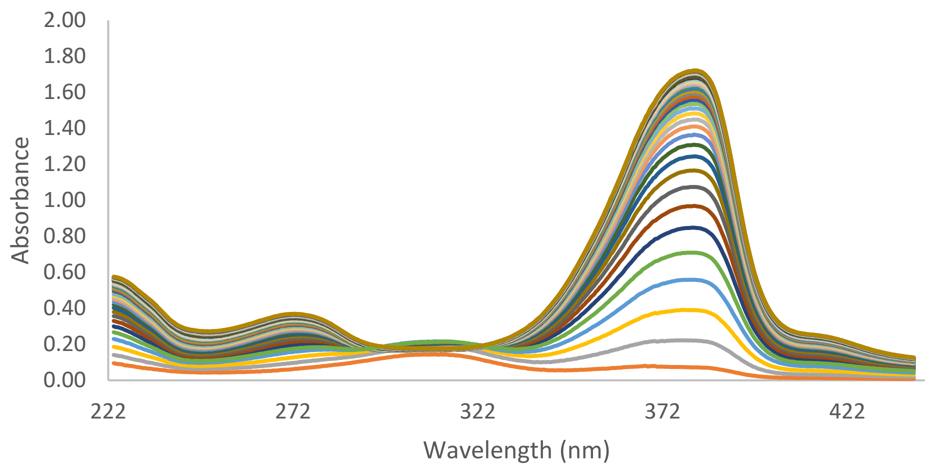


Figure 4. Time-series absorbance spectra at 60°C for the reaction of 0.01mg chenodeoxycholic acid with 74% w/w H2SO4. The observed maximum absorbance appears at approximately 380 nm.

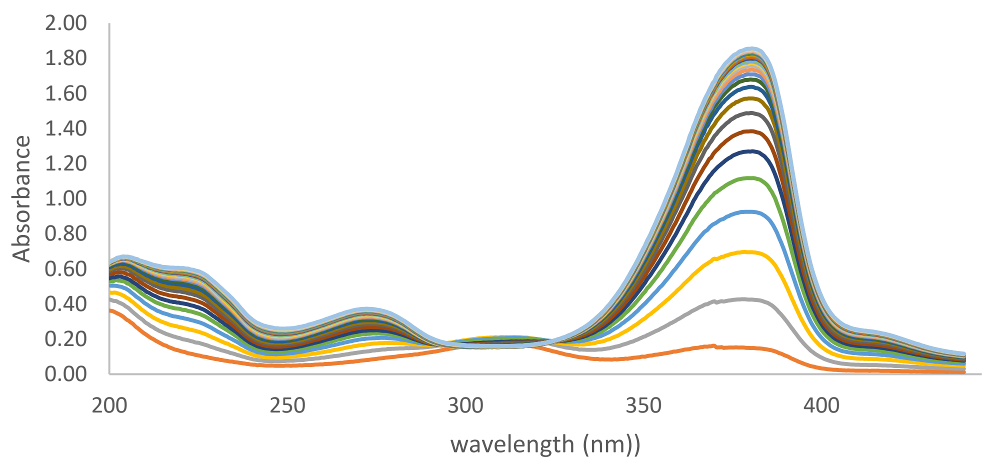


Figure 5. Time-series absorbance spectra at 65°C for the reaction of 0.01mg chenodeoxycholic acid with 74% w/w H2SO4. The observed maximum absorbance appears at approximately 380 nm.

Figures 6,7,8 and 9 shows the Absorbance verses time plot measured at 50,55,60 and 65°C. These constructed scatter plots indicate the different absorbances at three wavelengths:263, 310 and 380 nm.

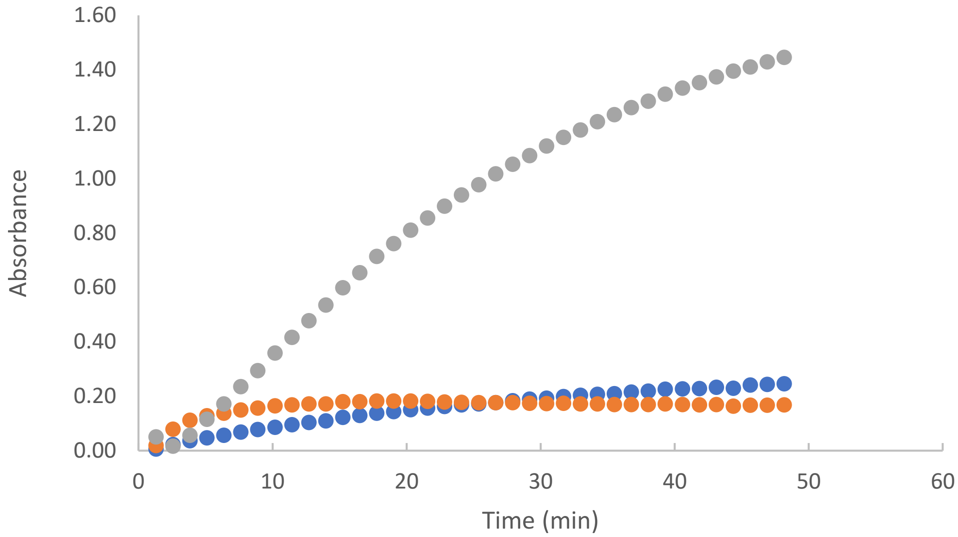


Figure 6. Absorbance vs. time plot measured at 50°C for the reaction of 0.01mg chenodeoxycholic acid with 74% w/w H2SO4. The scatter plots from three different wavelengths: 263,310 and 380 nm. Grey scatter plots represent 380nm, orange represents 310nm, and blue represents 263 nm.

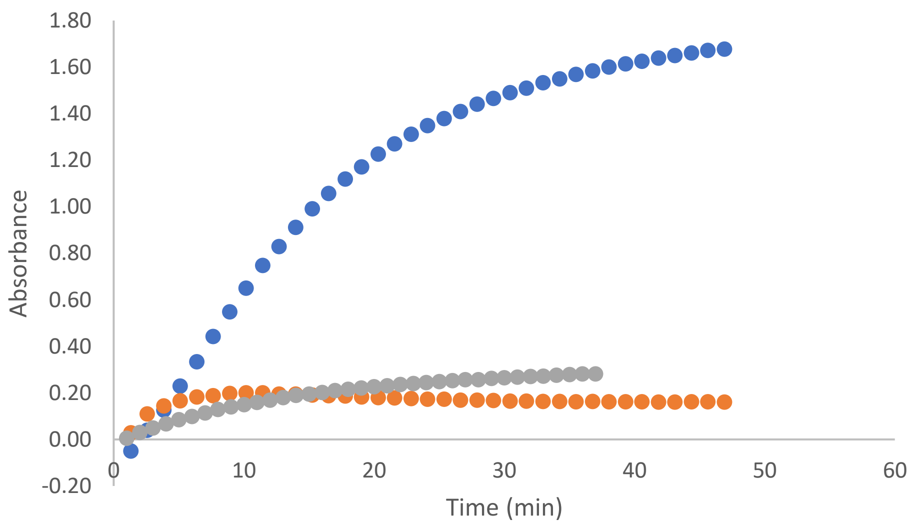


Figure 7. Absorbance vs. time plot measured at 55°C for the reaction of 0.01mg chenodeoxycholic acid with 74% w/w H2SO4. The scatter plots from three different wavelengths: 263,310 and 380 nm. Blue scatter plots represent 380nm, orange represents 310nm, and grey represents 263 nm.

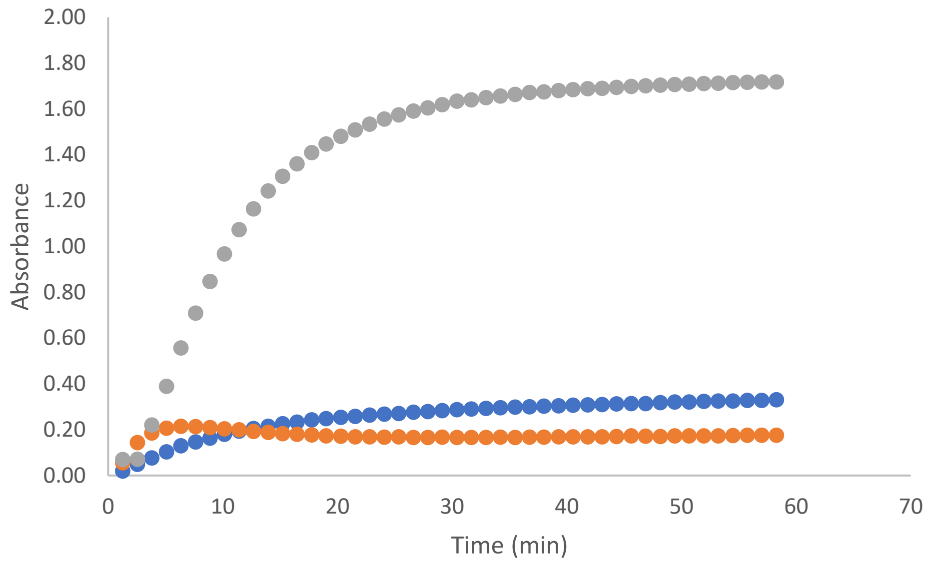


Figure 8. Absorbance vs. time plot measured at 60°C for the reaction of 0.01mg chenodeoxycholic acid with 74% w/w H2SO4. The scatter plots from three different wavelengths: 263,310 and 380 nm. Grey scatter plots represent 380nm, orange represents 310nm, and blue represents 263 nm.

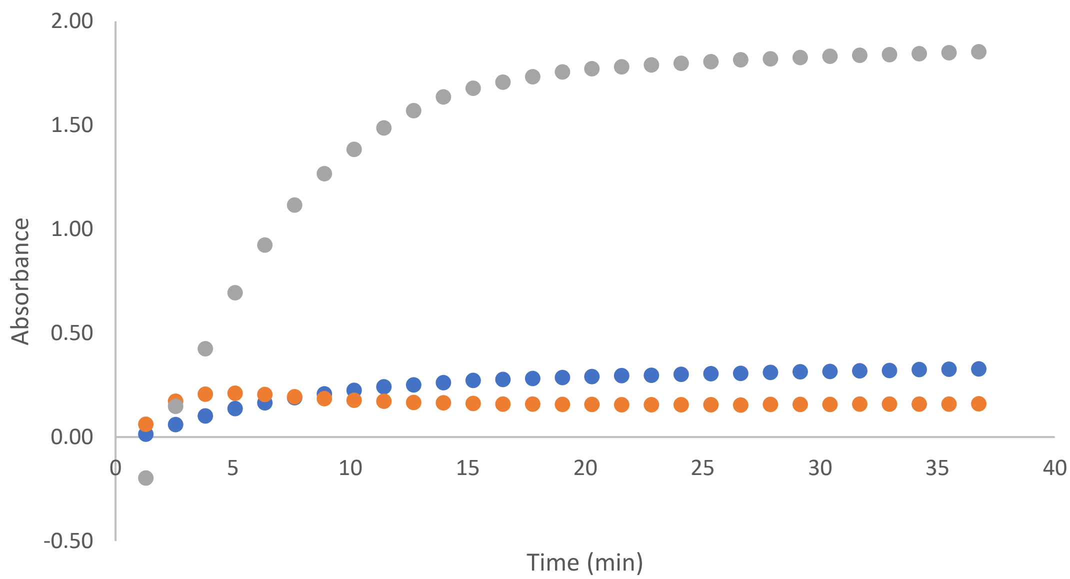


Figure 9. Absorbance vs. time plot measured at 65°C for the reaction of 0.01mg chenodeoxycholic acid with 74% w/w H2SO4. The scatter plots from three different wavelengths: 263,310 and 380 nm. Grey scatter plots represent 380nm, orange represents 310nm, and blue represents 263 nm.

Guggenheim treatment determines if the reaction is first or second order. The graphs show the data collected at various temperatures: 50,55,60 and 65°C. The linear behavior indicates first order reaction.

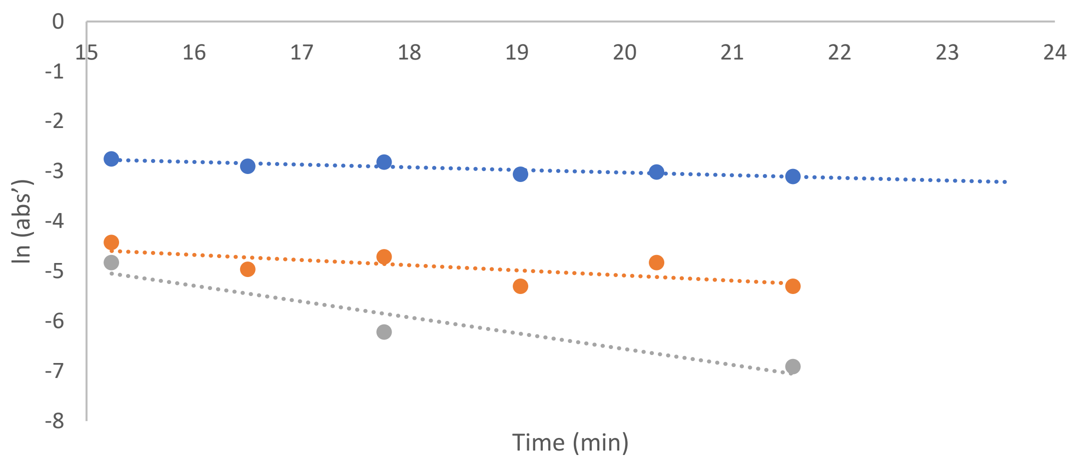


Figure 10. Guggenheim treatment at 50 °C with three different wavelengths: 263nm, 310nm, and 380nm. Linear behavior indicates first order reaction, and the slope of the linear data indicates the observed rate constant. Blue trendline indicates 380 nm, orange trendline is 263nm, and grey trendline indicates 310 nm. The average rate constant for this trail is 0.1576.

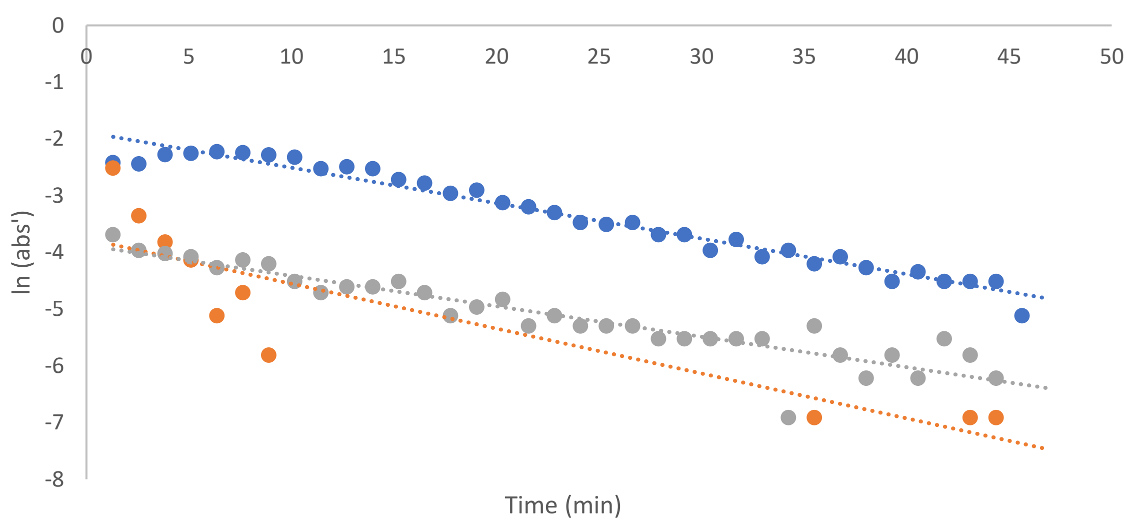


Figure 11. Guggenheim treatment at 55 °C with three different wavelengths: 263nm, 310nm, and 380nm. Linear behavior indicates first order reaction, and the slope of the linear data indicates the observed rate constant. Blue trendline indicates 380 nm, grey trendline is 263nm, and orange trendline indicates 310 nm. The average rate constant for this trail is 0.0651.

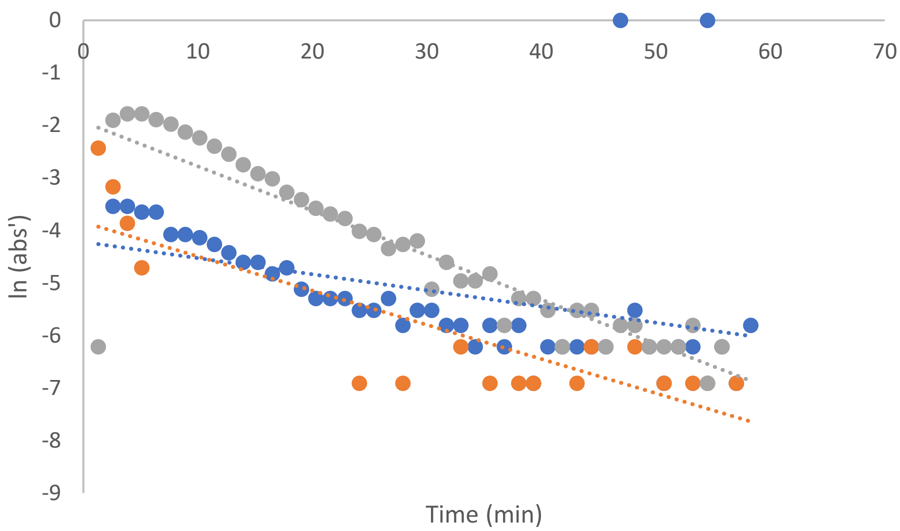


Figure 12. Guggenheim treatment at 60 °C with three different wavelengths: 263nm, 310nm, and 380nm. Linear behavior indicates first order reaction, and the slope of the linear data indicates the observed rate constant. Grey trendline indicates 380 nm, blue trendline is 263nm, and orange trendline indicates 310 nm. The average rate constant for this trail is 0.0601.

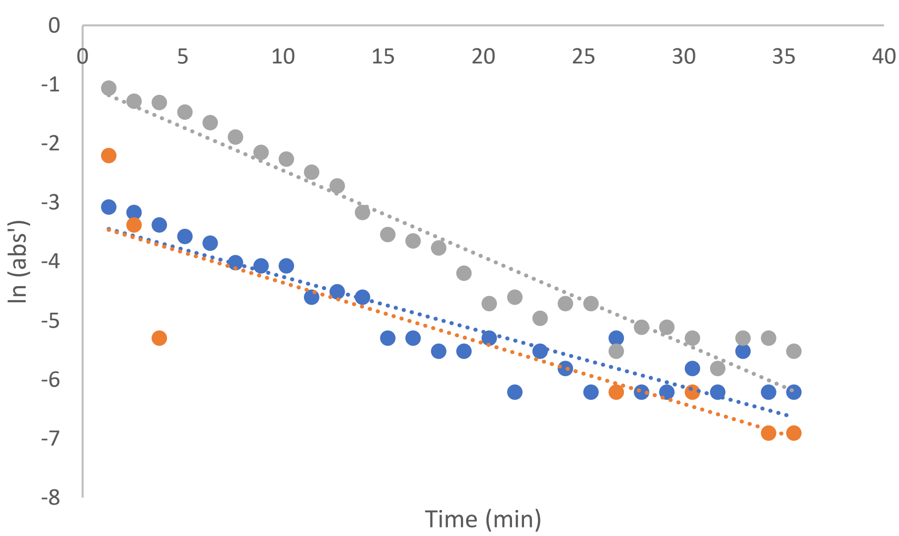


Figure 13. Guggenheim treatment at 65 °C with three different wavelengths: 263nm, 310nm, and 380nm. Linear behavior indicates first order reaction, and the slope of the linear data indicates the observed rate constant. Grey trendline indicates 380 nm, blue trendline is 263nm, and orange trendline indicates 310 nm. The average rate constant for this trail is 0.1141.

Table 1 shows the k-observed values at various temperatures for the reaction of 0.01 mg chenodeoxycholic acid with 74% w/w H_2SO_4. These variables are applied to the Arrhenius equation to find the activation energies at 263, 310, and 380 nm.

Table 1. Table of k-observed values at various temperature for the reaction of 0.01mg chenodeoxycholic acid

Temp	K-obs (avg)		1/T(k)	ln(k-obs)
50	263	0.034	0.003094538	-3.381
	310	0.0596		-2.820
	380	0.031		-3.474
55	263	0.057	0.003047387	-2.865
	310	0.0791		-2.537
	380	0.0625		-2.773
60	263	0.0307	0.003001651	-3.483
	310	0.0651		-2.732
	380	0.0846		-2.470
65	263	0.0932	0.002957267	-2.373
	310	0.0932		-2.373
	380	0.1465		-1.921

Figures 14,15 and 16 show the activation energies of chenodeoxycholic acid reaction with H_2SO_4 at 263nm, 310nm, and 380nm by using the Arrhenius equation. Activation energy at 263nm is 43.4 kJ/mol, 310nm is 20.8 kJ/mol, and 380nm is 90.2 kJ/mol.

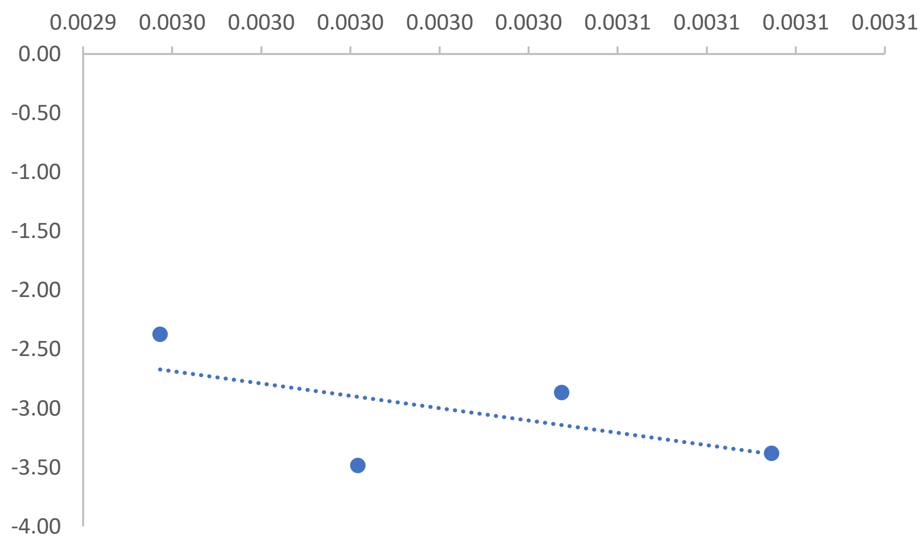


Figure 14. Arrhenius plot of all k-observed at 263 nm. ln(Abs)=-5219.9 (1/T) +12.766. Activation energy is 43.4 kJ/mol.

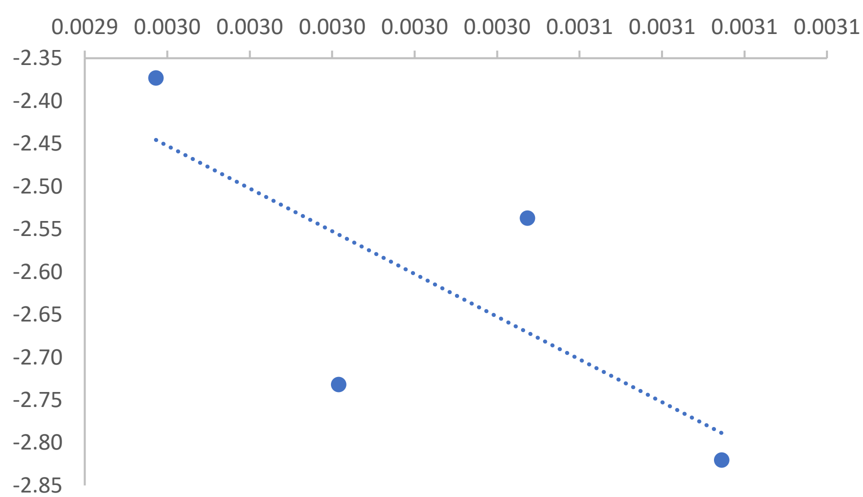


Figure 15. Arrhenius plot of all k-observed at 310 nm. ln (Abs)=-2500.6 (1/T) +4.9492. Activation energy is 20.8 kJ/mol.

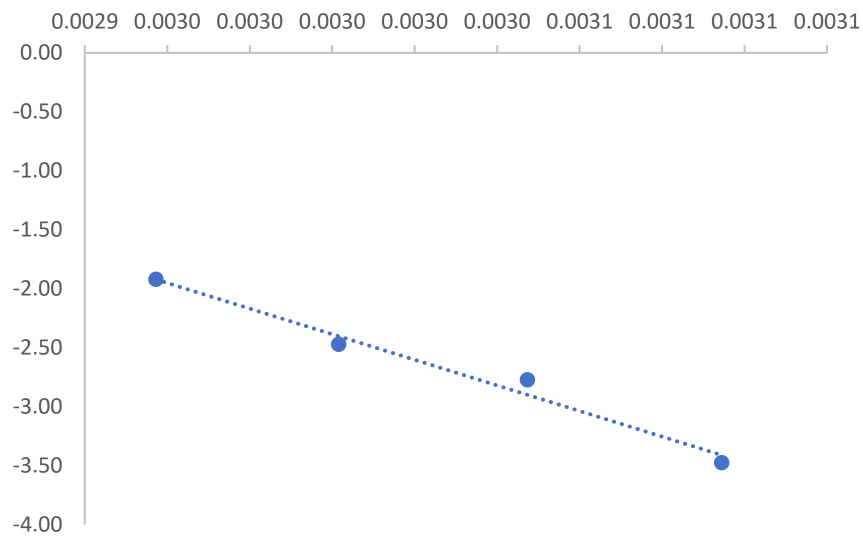


Figure 16. Arrhenius plot of all k-observed at 380 nm. ln(Abs)=-10853 (1/T) + 30.173. Activation energy is 90.2 kJ/mol.

Discussion

The figures 5,6 and 7 prove that the activation energies at three wavelengths are all different. At 263nm, the activation is 43.4 kJ/mol, 310 nm is 20.8kJ/mol, and 380 nm is 90.2 kJ/mol. The highest activation energy required compound has reaction at wavelength 380 nm, and the predicted mechanism proves that the compound at 380 nm is dineylic cation. Additional investigation to find the correlation between temperature and concentration of sulfuric acid with chenodeoxycholic acid may provide the resources to scientist to figure out what factors may decrease/increase the activation energy of chenodeoxycholic acid in our body.

Reference

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